

Finding the Hidden Product

Top-10 hit within 10 conversational turns · 50,000-item catalog · 0 external LLM calls

HIT@10

0.880

176 / 200 sessions · 7.0×

MRR

0.4916

7.2× official starter

MTTC

3.375

target found 2.9× faster

TECHNICAL SCORE

0.7400

7× starter (0.1067)

Architecture & the Simulator Contract

SIMULATOR CONTRACT (OFFICIAL EVALUATOR)

- 1 Every turn**
message + optional ask + up to 10 recommendations
- 2 Zero hit sacrifice**
a clarification question never forfeits the hit check
- 3 Defensive by default**
strict dedupe · valid catalog ASINs · spec-enum asks
- 4 Deterministic core**
offline execution · 332 automated tests guard the contracts

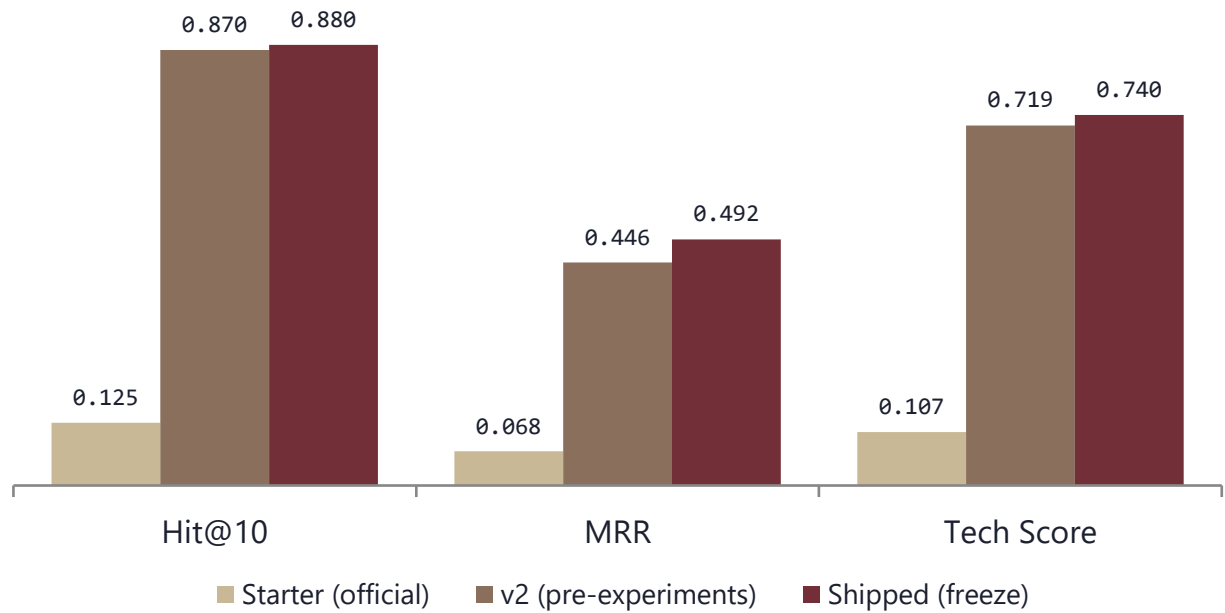
Evaluator imported unmodified — zero edits, imports only.

PRODUCTION EXECUTION FUNNEL

- 1. 50,000-item catalog**
SQLite FTS5 BM25 + hashed TF-IDF index
- 2. Hybrid pool · ~300 items**
Reciprocal Rank Fusion (k=60) + guarantee pool
- 3. Full scoring · 320 cap**
Cap $\geq$ pool $\rightarrow$ 100% of candidates scored
- 4. 8-gate clarification**
Entropy set-splitting on Boolean token index
- 5. Top-10, attributed**
Coverage $\times$ IDF $\times$ salience + popularity

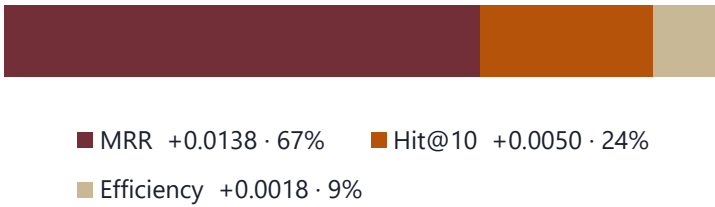
The Staircase: 7× the Official Baseline

OFFICIAL METRICS · PUBLIC-200



MTTC 9.81 → 3.38 turns — target found 2.9× faster

WHERE Δ +0.0207 CAME FROM



Two-thirds of the gain is MRR — the ranker puts the target higher in the top 10.

OFFICIAL FORMULA

$$\begin{aligned} \text{Score} &= 0.5 \cdot \text{Hit@10} \\ &+ 0.3 \cdot \text{MRR} \\ &+ 0.2 \cdot \text{clip}((11 - \text{MTTC}) / 10) \end{aligned}$$

Innovation Directions: What Shipped, What Died

SHIPPED — KEPT BY THE GATES

Hybrid retrieval

BM25 (0.7) + semantic TF-IDF (0.3) + RRF fusion

88% pool recall · 0.880 Hit@10

Constraint reweighting

buying-route salience 0.5 → 0.2, popularity untouched

+1.0pp Hit · +4.6pp MRR · SHIPPED

Explainable ranking

coverage · IDF · salience · popularity per item

live in demo · zero score penalty

KILLED — BY THEIR OWN PRE-REGISTERED BARS

LLM reranking tier

3 sizes & classes probed — 2.6B dense · 30B-A3B · 120B-A12B MoE + GPT-4o-mini · Hit / MRR ≈ flat

ΔHit 0 · ΔMRR -0.005 · +0.5-8.3 s per call → killed · local LLM loses on TTFT too

Soft personalization

profile boost, weight sweep 0.00 – 0.15

0.03 worsened MRR → set to 0.00

Late-phase question gate

margin-gain question value on misses

0 / 19 misses qualified → stopped

Method: Pre-Registered, Paired, Isolated

1

Worktree isolation

- 5 isolated experiment worktrees
- branched from control 80eee9a
- frozen evaluator · zero contamination

2

Pre-registered bars

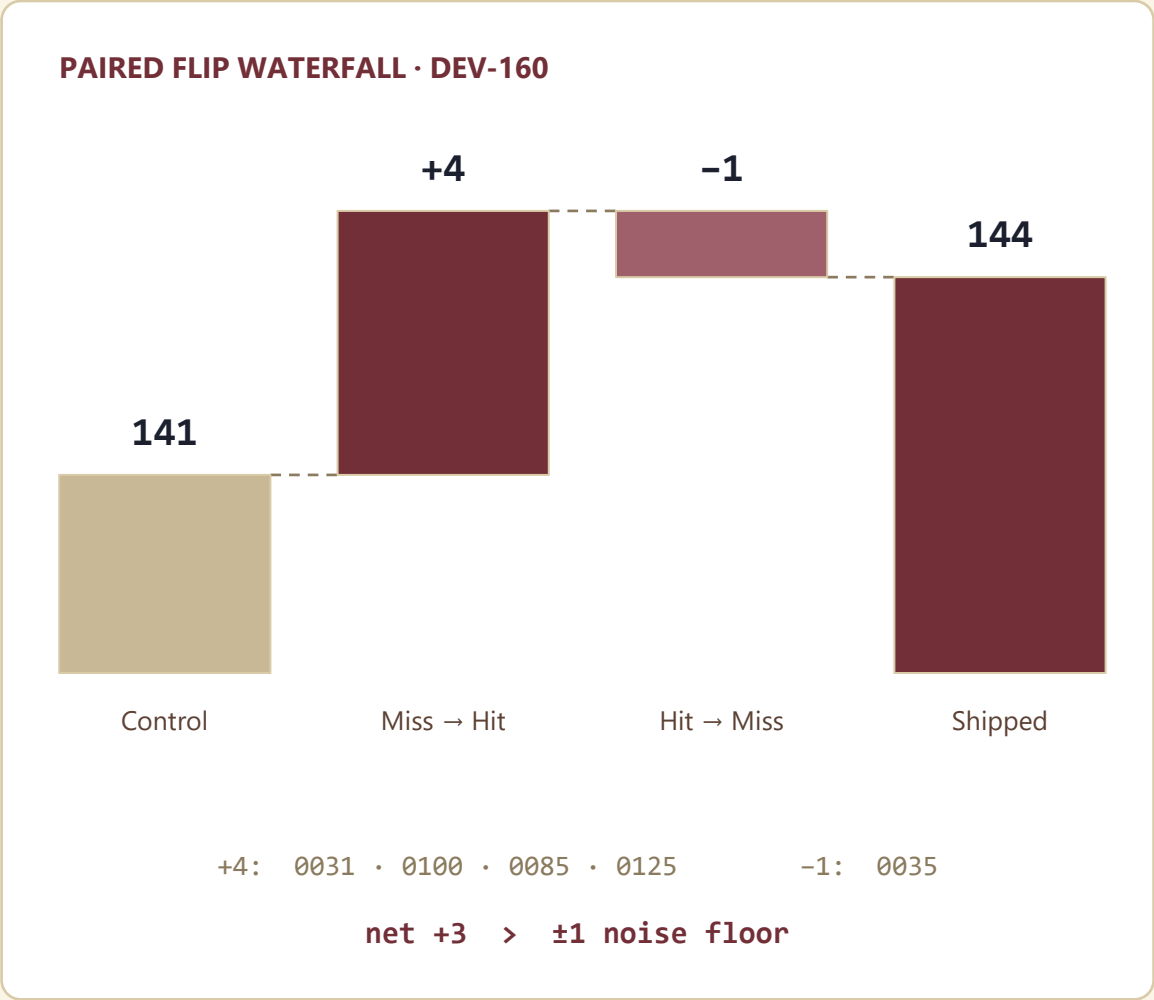
- pass bar written before first eval
- $\Delta\text{Hit} \geq +0.03$ · $p95 \leq +2$ s
- automatic revert on regression

3

Paired-session flips

- unit of truth: per-session flips
- noise floor ± 1 session (159/160 agree)
- aggregate shifts $< 3\text{pp}$ are noise

The Shipped Win: Salience vs Popularity



MECHANISM

- Popularity crowded out products satisfying rare, high-value constraints
- Fix: buying-route salience weight 0.5 → 0.2

HONESTY

Failed its own Buying-specific hypothesis — the wins were route-general. Shipped anyway: flips, not hypothesis, were the evidence.

public-200 confirm: Hit 0.870 → 0.880 · MRR
0.4455 → 0.4916

Out-of-Sample Confirmation

PROGRESSION ACROSS SPLITS

DEV-160 · TUNING SET

0.900 Hit@10

144 / 160 · MRR 0.5144 · MTTC 3.19

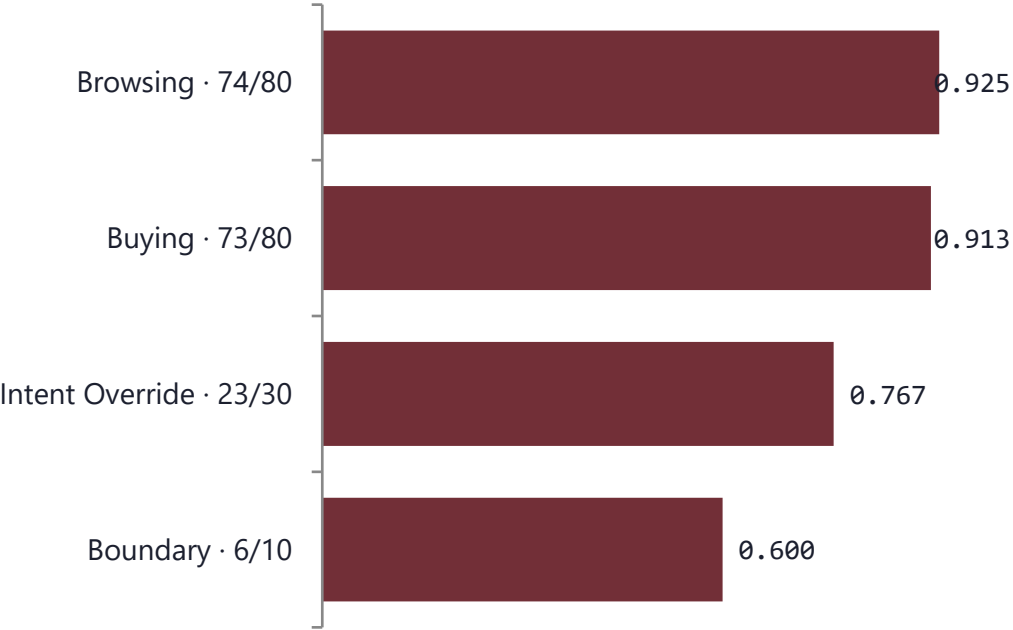
PUBLIC-200 · CONFIRMATION (TOUCHED 2×)

0.880 Hit@10

176 / 200 · MRR 0.4916 · MTTC 3.375

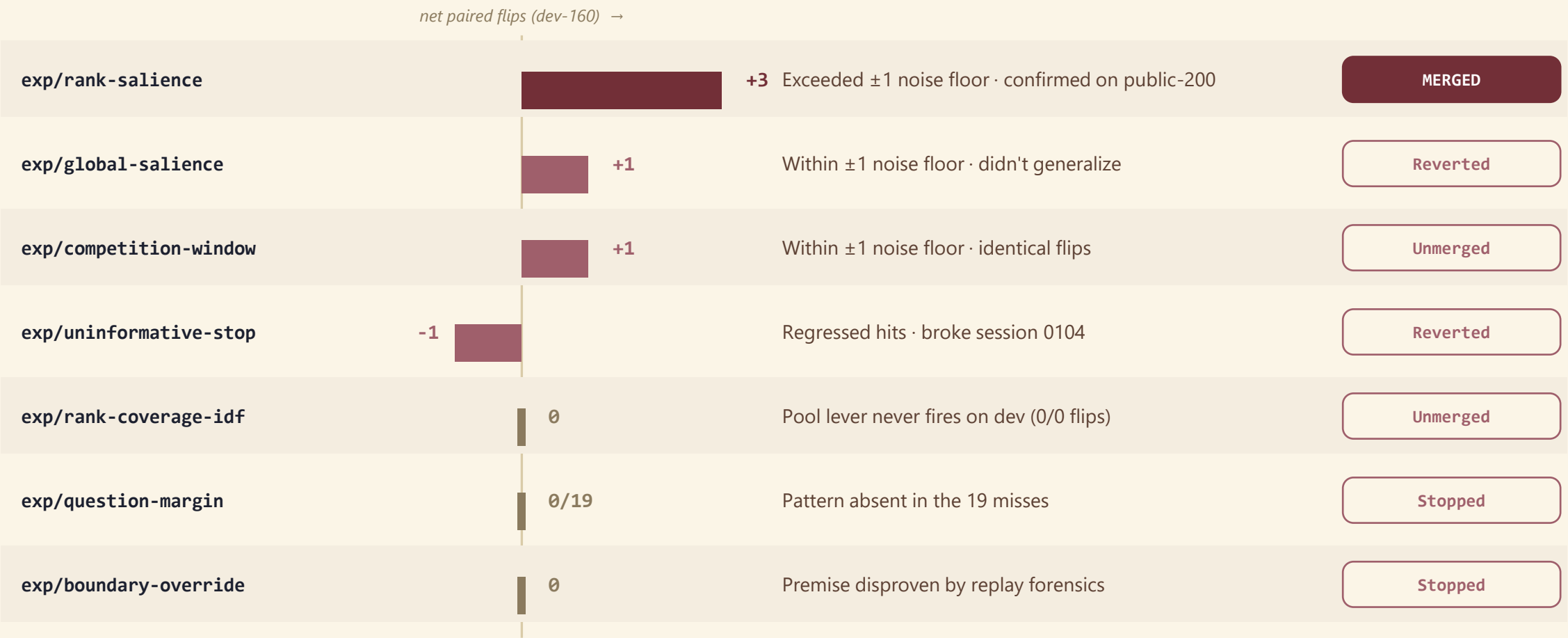
The salience gain was not overfitted to dev.

HIT@10 BY SCENARIO (PUBLIC-200)



Exact counts on every bar — small denominators stay visible.

The Graveyard: 6 Rejected by Their Own Bars



“Every red row was rejected by criteria written before the experiment ran. The 0.90625 in our logs was never merged.”

4 Structural Findings from Negative Results

01 MTTC = first-hit turn

miss = 11

The evaluator stops the moment the target enters top-10. Wasted questions happen post-hit — only surfacing sooner moves MTTC.

02 Dialogue and retrieval are coupled

0104

Stopping clarification changed the message stream and broke retrieval. “Hit-safe by construction” is false.

03 Scenario labels ≠ runtime route

89%

84 / 94 non-buying sessions run buying-route turns — including every one of the 15 dev misses.

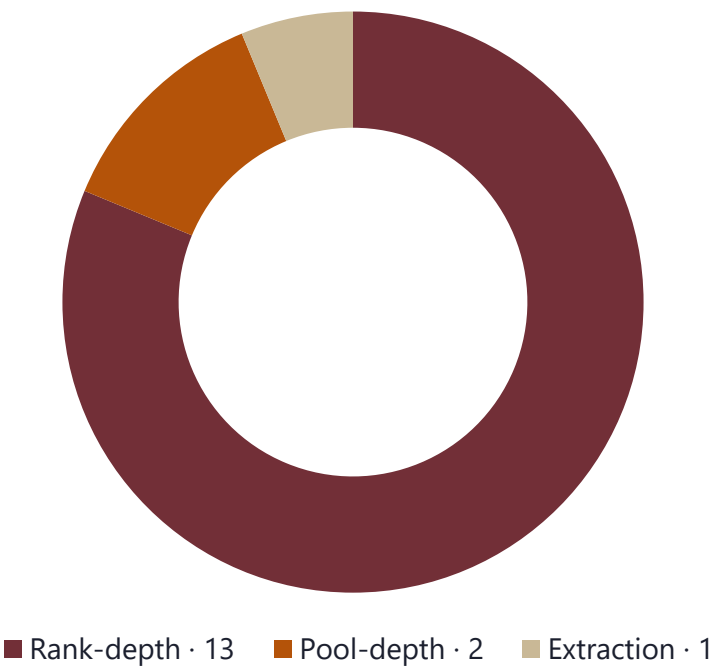
04 Re-ordering is saturated

+2/-1 ×2

Two independent ranker changes produced identical flips. Remaining misses need new information, not permutations.

Why We Miss: Forensic Audit of All 16 Dev Misses

MISS TAXONOMY (DEV-160)



n = 16 of 160 sessions

13

RANK-DEPTH

Target in the pool but outranked — needs new user information, not re-ordering.

2

POOL-DEPTH

Target ranked 201+ in the pool — a recall cap, not a filter bug.

1

EXTRACTION

Complex overlapping syntax edge case (public_0117).

QUERY FAITHFULNESS: 0 of 13 recoverable constraints dropped. The pipeline leaks nothing — remaining misses are genuine capability limits.

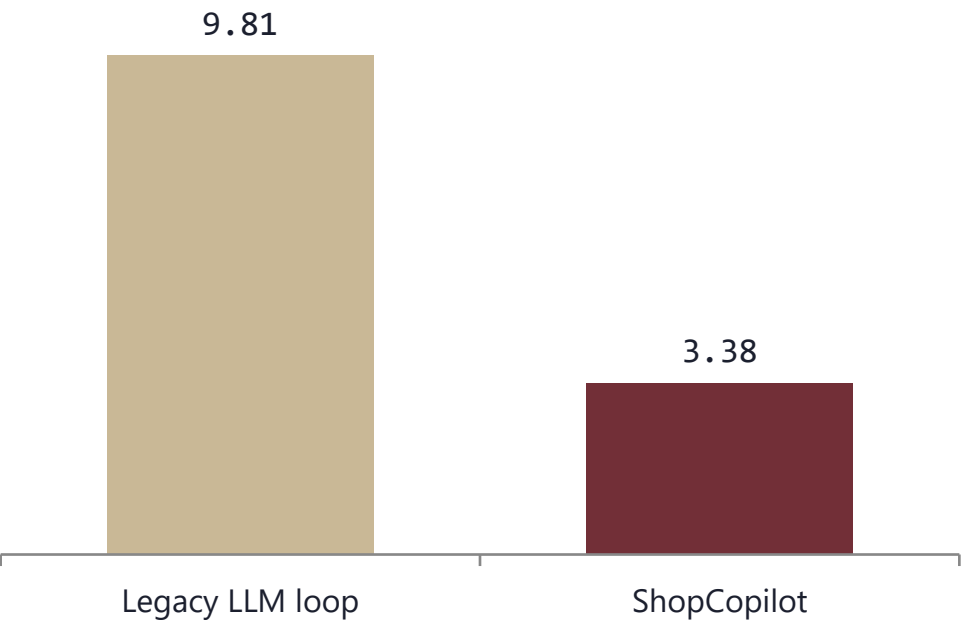
Feasibility: Zero Tokens, Zero Cost, Zero Network

LLM TOKENS 0 0 prompt · 0 completion	MODEL COST \$0.00 zero API fees	NETWORK None fully offline at inference	TURN LATENCY 330 ms p50 · p95 527 ms
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DISCLOSURE DETAIL	
Runtime model	Deterministic BM25 + hashed TF-IDF + constraint ranker – no model at inference
Evidence	results.json → reported_token_usage = 0 · total_tokens = 0
Optional LLM tier	Built and gated OFF by default (enable_llm_reranker = false)
Environment	Python 3.13 · Windows 11 Pro · AMD Ryzen 7 · 32 GB RAM
Recorded live	docs/final-eval-record.md – same frozen commit 46e3322

Business Impact: From Bounce to Checkout

Turns to find the product



Vague first query → wrong products → exit.

Bounce → checkout

3.4 turns vs 9.8 turns completes the purchase instead of abandonment.

Attributable ranking

Every card shows coverage, IDF rarity, salience, popularity — math, not a black box.

Zero marginal cost

330 ms responses, \$0.00 per turn, 100% uptime SLA — unit economics survive scale.

Final-Eval Readiness: 800 Hidden Sessions

✓ **Evaluator frozen**

byte-identical to starter package — imports only

✓ **Fresh-clone rehearsal**

clean clone reproduces the exact 0.880 table

✓ **No hardcoding**

zero session-id checks anywhere in the codebase

✓ **Data isolation**

800 hidden sessions touched 0 times · public-200 touched 2×

✓ **332 automated tests**

pass in 21 s on the frozen commit

✓ **Snapshot retained**

commit 46e3322 · pip deps · hardware captured

Roadmap: Attack the Measured Ceilings

PHASE 1

Recall expansion

Category-specialized inverted sub-indices break the rank-201+ candidate cap.

removes the 2 pool misses

PHASE 2

Deep feature enrichment

Extract fine-grained attributes — fabric weight, collar, sole stiffness.

feeds the 13 rank-depth misses

PHASE 3

Early hit prioritization

Optimize turn-1 ranking for top-1 placement, not just top-10 membership.

target: MTTC 3.38 → < 2.5

Provenance & Reproducibility

TAG CHAIN & KEY ARTIFACTS

```
fork-point      80eee9a  dev baseline
new-baseline    46e3322  post-saliency control
submission-freeze 46e3322  official submission
```

KEY ARTIFACTS

- results.json – public-200 · 0.880
- runs/control-dev-newbaseline.json – dev-160 · 0.900
- docs/experiment-ledger.md – every run
- DATA_ATTRIBUTION.md – UCSD McAuley Lab

REPRODUCE FROM A CLEAN CLONE

```
# 1 · install + build catalog index
pip install -r requirements.txt
python scripts/setup_catalog.py
```

```
# 2 · test suite (332 passed)
python -m pytest -q
```

```
# 3 · official evaluator → 0.880 table
python -m evaluator.local_evaluator
```

```
# 4 · live interactive demo
python scripts/interactive_demo.py
```